

Alan Siby

ajs9699@my.utexas.edu | Round Rock, TX | engrXiv Publication

Education

Round Rock High School

5.66 Weighted GPA / Coursework: Calculus, Physics, and other AP/Advanced courses

Aug 2022 – May 2026

Round Rock, TX

Austin Community College

Dual Enrollment / 4.0 GPA / Coursework: U.S. History, Macroeconomics, Government

Jun 2023 – May 2026

Austin, TX

Research Experience

Research Intern, Institute for Computing in Research

Jun 2025 – Jul 2025

- Designed and implemented a Python-based simulation of a 1-axis spacecraft attitude control system, modeling reaction wheel dynamics, actuator saturation, and proportional-derivative (PD) control
- Built a grid-search optimization routine to tune controller gains against settling time, overshoot, and control torque
- Diagnosed and corrected flawed performance-metric logic by systematically verifying simulation output against expected physical behavior, rather than trusting results that appeared numerically clean
- Co-authored findings and published on engrXiv, contributing to open-source aerospace controls research
- Collaborated with mentors and peers to refine algorithms and present results in professional technical format

Projects

1-Axis Spacecraft Attitude Control: PD Design with Reaction Wheel Saturation

engrXiv, 2025

- Built a fourth-order Runge-Kutta simulation framework in NumPy/Matplotlib to model closed-loop spacecraft dynamics under real actuator constraints
- Verified controller performance using root locus and time-domain analysis across six spacecraft configurations
- Identified and resolved a critical flaw in settling-time detection logic that produced false-positive stability results

Magic Snap: Ergonomic & Magnetic-Charging Shell for the Apple Magic Mouse

Team Flow State, 2026

- Co-designed and CAD-modeled a 3D-printed ergonomic shell addressing RSI risk and a magnetic charging dock resolving the device's original bottom-mounted charging flaw
- Directed user research and a weighted decision matrix across three competing team concepts to guide design direction
- Iterated through multiple prototypes, refining palm-arch fit, mouse glide, and charging-cable management based on user testing and interviews

Experience & Activities

Store Associate, The Home Depot

Mar 2026 – Present

- Help customers find plumbing products and answer questions about pipes, fittings, and fixtures
- Maintain a clean, organized, and safe work area while supporting team operations and customer service

Founder & Basketball Trainer, Apex Basketball Training

Sep 2024 – May 2026

- Founded and independently operate a youth basketball training business coaching 50+ athletes; manage scheduling, client communication, and marketing

NASA High School Aerospace Scholar

Oct 2024 – Jun 2025

- Selected for NASA's competitive national Aerospace Scholars program; collaborated on a team designing satellite enhancements for lunar exploration

DECA Competitor – Round Rock High School

Aug 2022 – May 2026

- Competed in business, marketing, and entrepreneurship events for four years; advanced to state competition

Awards

National Merit Commended (2025) | AP Scholar with Distinction (2025) | National Honor Society (2025) | AP Scholar (2024) | DECA SCDC State Qualifier (2025)

Skills

Programming: Python, Java

Tools: Autodesk Inventor (CAD), $\text{\LaTeX}$

Research: Technical writing, data analysis, control systems modeling